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Consequentialist Analysis of the Automation of Middle-Lower Class Labor:

Introduction

This paper examines the ethics of automating lower- and middle-class jobs through the framework of consequentialism. Specifically, it seeks to determine whether automating these jobs is ethical and, if not, how such automation might be carried out ethically.

The decision to focus on the automation of lower-skilled employment is motivated by two primary reasons. First, these jobs are among the most likely to be automated in the near future and therefore represent a particularly pressing ethical concern. As noted by the U.S. Government Accountability Office, "Workers with lower levels of education and who perform routine tasks, think cashiers or file clerks, face the greatest risks of their jobs being automated." Second, workers in these occupations tend to have limited occupational mobility, increasing their risk of long-term unemployment. This concern is supported by the article "Occupational Mobility and Automation: A Data-Driven Network Model," published by the Royal Society, which states, "In an automation scenario where low-wage occupations are more likely to be automated than high-wage occupations, the network effects are also more likely to increase the long-term unemployment of low-wage occupations."

For these reasons, the automation of lower- and middle-class labor is both a pressing and relevant issue in contemporary engineering practice. Engineers will be directly responsible for designing and implementing automated systems and must therefore contend with the ethical problems that arise from their deployment. While the ultimate decisions and policy solutions may not lie solely with engineers, policymakers will play a significant role, this is nevertheless an issue that engineering, as a professional practice, must address.

Ethical Framework

Consequentialism is an ethical framework that defines the morality of an action by its outcomes. Under this framework, an action is considered ethical if it produces the best overall consequences. More refined versions of consequentialist ethics also consider whether an agent has a reasonable expectation that their actions will lead to good outcomes. This distinction will be important later in the analysis.

The *Stanford Encyclopedia of Philosophy* describes consequentialism as follows: “This historically important and still popular theory embodies the basic intuition that what is best or right is whatever makes the world best in the future, because we cannot change the past, so worrying about the past is no more useful than crying over spilled milk.” Consequentialism is often the guiding ethical theory behind engineering decisions. A paper published by the College of Engineering and Science at Louisiana Tech University notes that “many engineering decisions are clearly based on a utilitarian approach, for most engineers assume that if an analysis method works in a given situation, then it is the proper choice.” Utilitarianism, a form of consequentialism, focuses on maximizing pleasure and minimizing pain.

A common ethical challenge arises when an action has a high probability of producing good outcomes but also a meaningful chance of producing harmful ones. In such cases, consequentialism requires careful evaluation. In this paper, Expected Utility Maximization (EUM) will be used as a tool to aid this analysis. The EUM equation is shown below:

$$EUM = (p_{good} \times U_{good}) + (p_{bad} \times -U_{bad})$$

p_{good} and p_{bad} would be the probability of the good and bad outcomes, with U_{good} and U_{bad} being the amount of good and bad. If the EUM value is negative, then the decision is unethical, conversely if the EUM value is positive then the decision is ethical. This formula, while not being definite, can certainly be helpful in our analysis.

With consequentialism established as the guiding ethical framework, it becomes clear why it is appropriate for analyzing the automation of labor. The amount of harm inflicted by mass unemployment must be carefully weighed against the potential benefits of automation in order to determine whether pursuing this path is ethically justified.

Case Study

Automation is not a new concept in engineering and has existed since the early twentieth century. According to *Merriam-Webster*, automation is defined as “the technique of making an apparatus, a process, or a system operate automatically.” In the context of the twenty-first century, automation primarily involves artificial intelligence systems performing administrative, analytical, and technical tasks, as well as robotic systems automating repetitive and operational work, such as manufacturing assembly and autonomous driving.

A common argument in engineering ethics is that the primary goal of an engineer is to create a better and easier life for the public, and automation often appears to serve this goal. However, as argued in the article “What Past Waves of Automation Can Teach Us About AI,” the current wave of automation differs significantly from those of the past. Historically, technology has tended to automate tasks within occupations rather than entire jobs. As the article notes, "Historically, it is less common for technology to fully automate an entire occupation."

The scope of jobs potentially affected by artificial intelligence and robotics presents a serious challenge for lower- and middle-class workers. A *Newsweek* article reports that consulting firm McKinsey estimates that up to 40 percent of jobs in the United States could be automated in the near future. The article also discusses a Senate report estimating that many lower- and middle-class jobs could be more than 50 percent automated. While jobs and industries have always evolved alongside technological progress, rarely has this transformation been so widespread or potentially comprehensive.

Although these developments may initially appear harmful, a proper ethical analysis requires evaluating whether automation is ultimately beneficial or harmful to society as a whole. There are potential advantages to automating these jobs, including increased productivity, economic growth, and the possibility of freeing individuals to pursue more meaningful or fulfilling work.

Consequentialist Argument for Automation

There are many consequentialist arguments in favor of automation, but one core argument appears repeatedly in the literature: that the rise of automation and artificial intelligence will usher in unprecedented levels of productivity and prosperity. In its article "Artificial Intelligence and the Future of Humans," the Pew Research Center quotes Erik Brynjolfsson, director of MIT's Initiative on the Digital Economy, who states, "I think it is more likely than not that we will use this power to make the world a better place. For instance, we can virtually eliminate global poverty, massively reduce disease and provide better education to almost everyone on the planet."

This expectation of large-scale benefit forms the strongest consequentialist justification for automation. While the outcomes are not guaranteed, proponents argue that there is a reasonable expectation that automation will improve conditions for the majority of people. In the same Pew Research Center article, Greg Shannon, chief scientist at Carnegie Mellon University's CERT Division, estimates that positive outcomes will outweigh negative ones by a significant margin. Shannon suggests that "better/worse will appear 4:1 with the long-term ratio 2:1," particularly noting that artificial intelligence performs well in repetitive tasks that humans often dislike.

The Pew Research Center article further reports that, of 979 expert respondents, 63 percent believe that most people will be better off as a result of artificial intelligence technologies. From a consequentialist standpoint, this expert consensus provides strong support for the ethical permissibility of automation. However, the remaining 37 percent of experts anticipate negative outcomes, raising important questions about the magnitude and distribution of potential harm. While probabilities can be estimated, the ethical analysis remains incomplete without considering the severity of negative consequences. This concern motivates an examination of the consequentialist arguments against automation.

Consequentialist Argument Against Automation

While the consequentialist argument for automation rests heavily on the promise of increased productivity, economic growth, and improved quality of life, these potential benefits must be weighed against the realistic and foreseeable harm that widespread automation presents. From a consequentialist standpoint, an action is ethical only if it produces the best overall outcomes. If the negative consequences are sufficiently severe or disproportionately concentrated among vulnerable populations, then even large aggregate benefits may fail to justify the action.

The most immediate and significant harm posed by automation is large-scale job displacement, particularly among lower- and middle-class workers. As discussed earlier, these workers are both more likely to be automated and less able to transition into new employment. This lack of occupational mobility significantly increases the probability of long-term or permanent unemployment. Unlike historical waves of automation that replaced specific tasks within jobs, AI-driven automation increasingly threatens entire occupations. When an occupation is automated rather than merely transformed, displaced workers cannot easily adapt by acquiring adjacent skills within the same field. From a consequentialist perspective, the prolonged unemployment and economic instability experienced by millions of workers represents a substantial amount of harm that must be factored into any ethical analysis.

Beyond unemployment itself, automation introduces secondary consequences that compound its negative impact. Job loss is strongly correlated with increases in mental health issues, substance abuse, family instability, and reduced life expectancy. These harms extend beyond the individual worker and affect families, communities, and social institutions. Lower-income communities, which are already more economically fragile, are likely to experience concentrated harm, leading to regional economic decline and increased dependence on public assistance. When viewed through a consequentialist lens, these cascading effects significantly increase the negative utility associated with automation.

When applying the Expected Utility Maximization (EUM) framework, the ethical concerns become even clearer. While proponents of automation assign a relatively high probability to positive outcomes, such as economic growth and improved living standards, the magnitude of the negative outcomes must also be considered. Long-term unemployment, loss of dignity associated with work, and systemic inequality represent extremely high negative utilities. Even if the probability of severe societal harm is lower than the probability of positive outcomes, the scale of that harm may outweigh the benefits. If the resulting EUM value is negative, then automation, as currently pursued, fails the consequentialist test for ethical action.

Taken together, these considerations suggest that the automation of lower- and middle-class labor, as it is currently unfolding, is ethically problematic from a consequentialist perspective. The foreseeable harms of mass unemployment, increased inequality, and widespread social disruption appear to outweigh the uncertain and unevenly distributed benefits. If engineers and policymakers continue to pursue automation without addressing these consequences, they risk violating the core consequentialist principle of maximizing overall well-being.

Ethical Solutions and Consequentialist Evaluation

From a consequentialist perspective, the ethical permissibility of automation does not depend solely on whether automation occurs, but on how it is implemented and how its consequences are distributed across society. Several distinct approaches to automation are available, each with different expected outcomes. This section evaluates three ethically relevant options using a consequentialist framework and EUM formula.

Option 1: Unrestricted Market-Driven Automation

The first option is to allow automation to proceed largely unchecked, guided primarily by market incentives such as cost reduction, productivity gains, and competitive advantage. Under this approach, engineers focus on technical optimization, while social consequences such as job displacement are left to be addressed indirectly by market forces.

From a consequentialist standpoint, this option offers significant potential benefits. Increased productivity and efficiency could lead to economic growth, lower consumer prices, and technological innovation. These benefits generate positive utility for businesses, consumers, and high-skill workers who are able to adapt to new roles.

However, the negative consequences of this option are both foreseeable and severe. Large-scale displacement of lower- and middle-class workers would likely result in long-term unemployment, economic insecurity, and widening inequality. As discussed earlier, these workers have limited occupational mobility, meaning that the harms are not temporary but persistent. Secondary effects—such as increased mental health crises, community decline, and strain on public welfare systems—further increase negative utility.

When evaluated using Expected Utility Maximization, the magnitude of these harms weighs heavily against the benefits. Even if the probability of economic growth is high, the scale and concentration of suffering among vulnerable populations produce a large negative utility. As a result, unrestricted automation performs poorly under consequentialist analysis and fails to maximize overall well-being.

Option 2: Automation with Policy-Based Mitigation

A second option is to continue advancing automation while implementing policy interventions designed to mitigate its harm. These interventions may include retraining programs, expanded unemployment benefits, wage insurance, and public investment in job transition pathways. Engineers continue to develop automated systems, but deployment occurs alongside social safeguards.

Consequentially, this option improves unrestricted automation by reducing the severity and duration of harm experienced by displaced workers. Retraining and transitional support increase occupational mobility, lowering the probability of long-term unemployment. This reduces negative utility while preserving many of automation’s economic benefits.

However, this option still presents ethical challenges. Retraining programs are not universally effective, particularly for older workers or those in highly specialized roles. Additionally, policy interventions are often unevenly implemented and politically vulnerable. If safeguards are insufficient or delayed, significant harm may still occur. From a consequentialist perspective, this introduces uncertainty into the expected outcomes.

Under EUM analysis, this option yields a higher expected utility than unrestricted automation, as both the probability and magnitude of harm are reduced. However, because the effectiveness of mitigation efforts is not guaranteed, this approach still risks producing net negative outcomes if implementation fails. Thus, while ethically preferable to Option 1, it remains morally fragile.

Option 3: Human-Centered Automation and Augmentation

The third option prioritizes human-centered automation, in which engineers deliberately design systems to augment human labor rather than replace it wherever possible. Automation is deployed selectively, with explicit consideration of its impact on employment, and replacement is pursued only when strong justifications exist and adequate safeguards are in place.

From a consequentialist standpoint, this option directly targets the largest sources of negative utility associated with automation. By preserving employment and worker dignity, it minimizes the risk of long-term unemployment and its cascading social harms. At the same time, augmentation-based automation still allows for productivity gains, improved safety, and reduced engagement in undesirable tasks.

While this approach may produce slower economic gains and higher short-term costs, the long-term consequences are more evenly distributed and socially stable. The reduction in severe harm to vulnerable populations significantly lowers negative utility. When evaluated using EUM, this option produces the highest expected utility because it combines moderate positive outcomes with a substantial reduction in catastrophic negative consequences.

Consequentially, this option best satisfies the ethical requirement to maximize overall well-being. By intentionally shaping technological development to reduce foreseeable harm, engineers act in accordance with both professional responsibility and ethical obligation.

Conclusion

This paper sets out to examine the ethics of automating lower- and middle-class labor through the framework of consequentialism. By analyzing both the arguments for and against automation, it becomes clear that the ethical status of automation hinges not on its technical feasibility or economic efficiency, but on its real-world consequences for society as a whole.

On the one hand, automation promises significant benefits. Increased productivity, economic growth, and the potential to reduce human involvement in undesirable or dangerous work represent genuine positive outcomes. Many experts reasonably expect that AI and automation could improve living standards for a majority of people. These arguments form the strongest consequentialist case for continuing to develop and deploy automation technologies.

On the other hand, the harms associated with automation are substantial and well-supported by current evidence. Job displacement disproportionately affects lower- and middle-class workers who lack the mobility and resources needed to adapt. The resulting unemployment and economic insecurity generate secondary harms that ripple through families, communities, and society at large. Moreover, automation threatens to exacerbate existing inequalities by concentrating wealth and power while eroding the economic stability of those most vulnerable. As one analysis notes, if the distribution of wealth is left to free-market forces alone, the outcome is likely to be deeply unjust, a trend that is already becoming visible.

From a consequentialist standpoint, these harms cannot be dismissed as temporary or incidental. When evaluated using Expected Utility Maximization, the magnitude of the negative outcomes casts serious doubt on whether automation, in its current form, produces a net positive result. Even if the probability of positive outcomes is relatively high, the severity and scale of the potential harms may outweigh those benefits, rendering the practice ethically impermissible.

Ultimately, this analysis suggests that automation is not inherently unethical, but its current trajectory is deeply concerning. For automation to be ethically justified under consequentialism, significant changes must be made to how its consequences are managed. This includes robust policy interventions, meaningful retraining programs, and deliberate efforts to ensure that the benefits of automation are distributed more equitably. Without such measures, continuing to automate lower- and middle-class labor risks causing more harm than good, failing the very ethical framework that offers automation its strongest defense.

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